SIMATS ENGINEERING

ASSESSMENT TOOL 2 : Scenario-Based

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1520

COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments.* (BL3)

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

Total Marks: 25

Code of Conduct

I Kenisha Judith J (192572071) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Kenj

Assessment Tasks

Scenario-Based Questions

SET 2:

1.A retail company collects terabytes of customer transaction data every day from online and offline stores. The management wants to analyze purchasing patterns in near real time to improve product recommendations and inventory management.

- Design a cloud-based big data architecture using distributed storage and processing frameworks.
- Explain how batch processing and stream processing can be integrated.
- Discuss how data lakes and data warehouses can be used together.
- Recommend techniques to ensure scalability and cost efficiency.

2.A healthcare organization is migrating patient records and medical imaging data to the cloud. The solution must comply with strict privacy regulations while allowing authorized doctors to access records securely from multiple locations.

- Design a secure cloud architecture that protects sensitive healthcare data.
- Explain how encryption, identity and access management (IAM), and key management should be implemented.
- Discuss auditing, logging, and compliance monitoring strategies.
- Recommend methods to prevent unauthorized access and data breaches.

3.An online ticket booking platform experiences sudden spikes in traffic whenever popular events are announced. The company wants to reduce infrastructure management while ensuring fast response times.

- Design a serverless cloud architecture for the application.
- Explain how event-driven computing can improve scalability.
- Discuss the role of managed databases, API gateways, and message queues.
- Evaluate the advantages and limitations of serverless computing for this workload.

4.A multinational company gathers data from sales, finance, manufacturing, and customer support systems. The organization requires a centralized analytics platform with high-quality and consistent data.

- Design a cloud-based ETL/ELT pipeline for integrating data from multiple sources.
- Explain how data quality, metadata management, and governance should be maintained.
- Discuss the role of workflow orchestration and scheduling tools.
- Recommend strategies for optimizing query performance and storage costs.

5.A logistics company wants to predict delivery delays using historical shipment data, weather information, and live GPS tracking. The analytics platform must support continuous model training and real-time predictions.

- Design a cloud-based machine learning and analytics architecture.
- Explain how big data technologies can support model training and inference.
- Discuss the use of cloud-native AI/ML services for model deployment and monitoring.
- Recommend strategies for handling large-scale data ingestion, feature engineering, and model updates.

1) Introduction :

A retail company collects large volumes of transaction data from online and offline stores. A cloud-based big data architecture can process this data efficiently to identify purchasing patterns, improve product recommendations, and optimize inventory management.

1. cloud - Based big data Architecture

The architecture consists of the following components :

* Data ingestion : Use Apache Kafka or cloud-based streaming services to collect transaction data from stores, websites and mobile applications.
* Distributed storage : store raw and processed data in a cloud data lake using scalable object storage.
* Distributed processing : Use Apache spark or similar frameworks to process large datasets across multiple computing nodes.
* Analytics and reporting : use a cloud data warehouse and business intelligence tools to analyze purchasing trends and generate reports.

2. Integrating of Batch and stream Processing

Batch processing analyzes large amounts of

historical transaction data at scheduled intervals. It helps identify long-term purchasing trends, seasonal demand, and customer preferences.

stream processing analyzes incoming transactions continuously. It enables the company to update product recommendations and detect changes in demand quickly.

combining both methods provides historical insights and near real-time analytics.

3. Data Lakes and Data Warehouses:

A data lake stores raw data in different formats, including structured transaction records and semi-structured application logs. A data warehouse stores cleaned, structured and organized data for reporting and business analysis.

The company can use the data lake for large-scale storage and processing, then transfer validated data into the warehouse for dashboards, sales reports and inventory analysis

4. Scalability and cost efficiency

* use auto-scaling to increase or decrease computing resources according to workload.
* Apply data partitioning and compression to improve processing efficiency.
* use storage lifecycle policies to move older data to lower-cost storage.
* use serverless analytics or managed processing services where scalable.
* Monitor resource consumption and shut down unused computing resources.

conclusion:

* cloud-based big data architecture combining distributed storage, batch processing, stream processing, and data warehousing enables the retail company to analyze data efficiently. It improves customer recommendations, supports inventory planning and maintains scalability while controlling costs.

2) Introduction:

Healthcare organizations store sensitive information, including patient records, medical images, and treatment histories. A secure cloud architecture must protect confidentiality,

maintain data integrity, and allow authorized doctors to access information from different locations while meeting applicable privacy regulations.

+ secure cloud Architecture

The architecture should include the following components:

* cloud storage: store electronic health records and medical images in encrypted cloud databases and object storage.

* secure network: use private networks, VPNs, and appropriate network security rules to protect communication between hospitals and cloud services.

* Application layer: Provide a secure healthcare application through which authorized doctors can access patient information.

* Identity management: use centralized identity and access management to verify users and control their permissions.

* Backup and recovery: Maintain protected backups and recovery procedures to prevent data loss.

2. Encryption, IAM and key management

Encryption: Encrypt patient records and medical images at rest and use TLS encryption for data transmitted between hospitals, doctors and cloud services. This prevents unauthorized parties from easily reading intercepted or stolen data.

Identity and Access Management (IAM): Implement role-based access control. Doctors should access records required for patient treatment, while administrators should receive only the permissions necessary for their duties. Multifactor authentication (MFA) should be enabled for sensitive access.

Key management: use a cloud key management service to securely create, store, rotate and control encryption keys. Access to keys must be restricted and audited separately from access to patient data.

3. Auditing, Logging and compliance Monitoring

Maintain audit logs for patient-record access, modifications, downloads; authentication attempts, and administrative activities. Use centralized logging and monitoring tools

to identify unusual access patterns and generate security alerts.

conduct regular security assessments, review permissions and monitor compliance with applicable healthcare privacy and data - protection regulations. Audit logs should be protected from unauthorized alteration.

+ Preventing Unauthorized Access and data Breaches

★ Apply least - privilege access and regularly review user permissions.

★ Enable MFA and strong authentication policies.

★ use firewalls, network segmentation, and intrusion detection.

★ Encrypt sensitive data and maintain secure backups.

★ Patch applications and cloud systems regularly.

★ Train healthcare employees to recognize phishing and other security threats.

conclusion :

A secure cloud architecture using encryption,

IAM, key management, auditing and continuous monitoring protects sensitive healthcare information. These controls allow authorized doctors to access records securely while reducing the risk of data breaches and supporting regulatory compliance.

3) Introduction:

An online ticket booking platform may experience sudden increases in traffic when popular events are announced. A serverless architecture helps manage these fluctations without requiring the company to maintain and manage physical servers or continuously provision fixed computing capacity.

1. serverless cloud Architecture

The proposed architecture includes:

★ API gateway: Receives booking requests from web and mobile applications and routes them to the appropriate backend services.

★ serverless functions: services such as AWS Lambda or Azure functions process ticket searches, booking requests, payments and confirmations.

* Managed database : stores event details, customer information, booking records and ticket availability.

* Message queue : Temporarily holds booking tasks during heavy traffic and allows them to be processed reliably.

* Monitoring service : Tracks errors, response times, request volumes and resource usage.

2. Role of Event-Driven Computing

Event-driven computing executes functions in response to events such as a booking request, payment confirmation or ticket availability update.

When traffic increases, the serverless platform can automatically run additional function instances within its configured limits.

Message queues help absorb sudden request spikes and distribute work to backend services.

This improves scalability and prevents every request from directly overloading the database.

3. Managed databases, API Gateways and Message Queues

Managed databases: Reduce database administration by allowing the cloud provider to handle much of the maintenance, backup and infrastructure management. Transactions, concurrency controls and appropriate database constraints are essential to prevent the same ticket from being sold more than once.

API gateways: Provide a controlled entry point for requests and support authentication, request routing, throttling and rate limiting

Message queues: separate incoming requests from backend processing. They help manage traffic spikes, support retries and reduce the impact of temporary service failures.

4. Advantages and limitations of serverless computing

Advantages:

★ Automatically scales according to incoming requests, within service limits.

★ Reduce server maintenance and infrastructure administration

★ charges are often based on actual execution and resource consumption.

* supports rapid development and deployment

Limitations :

* cold starts may increase response times for some functions.
* Execution duration, concurrency and other service limits may apply.
* Heavy workloads can create unexpected costs.
* Debugging distributed functions and monitoring failures can be complex.

conclusion :

A serverless architecture using API gateways, functions, managed databases and message queues can handle sudden ticket-booking traffic effectively. Database transaction controls, request throttling, queue management and monitoring are necessary to maintain fast response times and prevent duplicate bookings.

4) Introduction:

A multinational company generates data from multiple departments, each potentially using different formats, systems and data standards. A cloud-based ETL/ELT pipeline integrates these sources into a centralized analytics platform.

1. cloud-based ETL/ELT pipline

The proposed pipline contains the following stages:

1. Data extraction: collect data from sales applications, finance systems, manufacturing databases and customer support platforms using APIs, database connectors or scheduled exports.

2. Data ingestion: Transfer the collected data into cloud object storage or a staging area.

3. Data transformation: clean, validate, standardize, and combine the data using cloud data integration tools or distributed processing frameworks.

4. Data loading: store the processed data in a cloud data warehouse for reporting and

analysis.

5. Virtualization: connect business intelligence tools to the warehouse to create dashboards and reports.

In ETL, data is transformed before loading into the target system. In ELT, raw data is loaded first and transformed within the target data platform.

2. Data quality, Metadata and governance

Data quality: Apply validation rules to identify missing values, duplicate records, invalid dates, and consistent formats. standardize currencies, product codes, customer identifiers and time zone where necessary.

Meta data management: Maintain information about data sources, formats, owners, transformations and update schedules. A data catalog helps user discover and understand available datasets.

Data governance: Establish data ownership, access permissions, classification rules,

retention policies, and quality standards: sensitive financial and customer information should be accessible only to authorized users.

3. Workflow Orchestration and scheduling

Workflow orchestration tools coordinate the execution of pipeline tasks and manage dependencies between them. scheduling tools run jobs at appropriate intervals, while monitoring identifies failed tasks and delayed data.

For example, the pipeline can load daily sales data, validate financial records, combine manufacturing information and refresh reporting tables. Failed tasks should ~~reexecute~~ support controlled retries and alters to administrators.

4. Query performance and storage cost optimization

★ Partition large datasets by date or other frequently used filtering fields.

* use columnar storage formats and compression where appropriate.
* Apply indexing, clustering or materialized views when supported by the platform.
* Remove unnecessary duplicate and temporary data.
* use lifecycle policies to move older raw data to lower-cost storage.
* optimize queries and monitor warehouse computing consumption.
* scale computing resources according to workload requirements.

conclusion:

A cloud-based ETL/ELT pipeline integrates departmental data into a centralized analytics platform. Data validation, metadata management, governance and workflow orchestration and lifecycle management reduce processing time and storage costs.

5) Introduction :

A logistics company can use machine learning to predict delivery delays by analyzing historical shipment records, weather conditions, traffic-related information and live GPS tracking. A cloud-based architecture provides scalable data ingestion, distributed processing, model training, and real-time predictions.

1. Cloud-Based Machine Learning Architecture

The architecture consists of the following components:

* Data integration: collect historical shipment records, weather updates and GPS data using APIs, streaming services and message brokers.
* Cloud storage: store raw data in a cloud data lake and maintain processed datasets for analytics.
* Data processing: use distributed processing tools such as Apache spark to clean, combine and process large datasets.

- ★ Feature engineering: create useful features such as distance remaining, average vehicle speed, weather severity, historical route delays, and estimated arrival time.

★ Model training: use cloud-machine learning services to train a model on historical delivery outcomes.

★ Model deployment: Deploy the trained model as an online prediction endpoint or managed interference service.

★ Monitoring: Track prediction accuracy, system performance and changes in incoming data.

2. Big Data Technologies for Training and Interference

Big Data technologies helps processing large volumes of shipment records and continuous GPS updates. Distributed framework such as Apache Spark can clean and combine data across multiple computing nodes, making large scale feature preparation and model training more efficient.

Historical data is used to train the model to identify patterns associated with delayed deliveries. During interference, current GPS coordinates, weather conditions and shipment details are provided to the deployed model to estimate the likelihood or expected duration of a delay.

streaming technologies can process incoming GPS updates continuously, enabling predictions to be refreshed as delivery condition change.

3. Cloud-Native AI/ML Services for Deployment and Monitoring

Managed cloud AI/ML services simplify model training, deployment, versioning and monitoring the model can be deployed through an API endpoint that logistics applications use to request delivery predictions.

Monitoring tools should track model accuracy, prediction, latency, errors and changes in data distributions. If prediction quality decreases, the organization can investigate the cause and retain or replace the model.

4. Large-Scale Data Ingestion, Feature Engineering and Model Updates

* use streaming services or message brokers to handle continuous GPS and weather data
* Apply scalable storage and partitioning to organize shipment records efficiently.
* Validate incoming data and remove duplicates or incorrect GPS readings.
* Generate features consistently during both training and real-time prediction.

conclusion:

A cloud-based machine learning architecture combines scalable data ingestion, distributed processing, feature engineering, model training and real-time interference. cloud-native AI/ML services simplify deployment and monitoring, while automated data pipelines and model updates help maintain prediction accuracy.